DRIVING IN THE WRONG DIRECTION?

A CO-EVOLUTIONARY MODEL OF ELECTRIC VEHICLE ADOPTION AND INNOVATION

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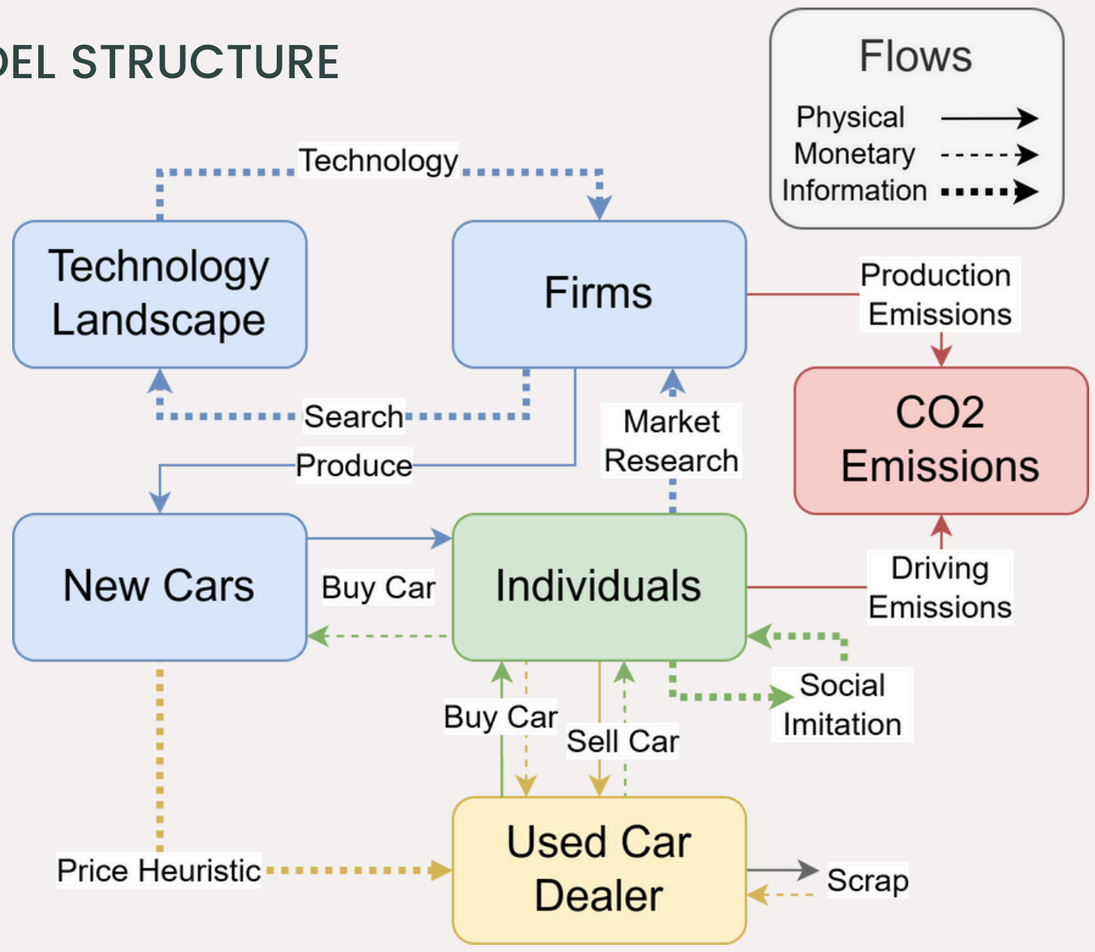
INTRODUCTION

Car-centric societies face substantial challenges in moving towards sustainable mobility systems, with internal combustion engine vehicles remaining a major source of emissions. Electric vehicles play a critical role in addressing this challenge, yet their diffusion depends on the interaction of consumer behaviour, firm innovation, and policy incentives.This paper develops an Agent-based Model (ABM) to examine these dynamics and guide effective climate policy.

RESEARCH QUESTION

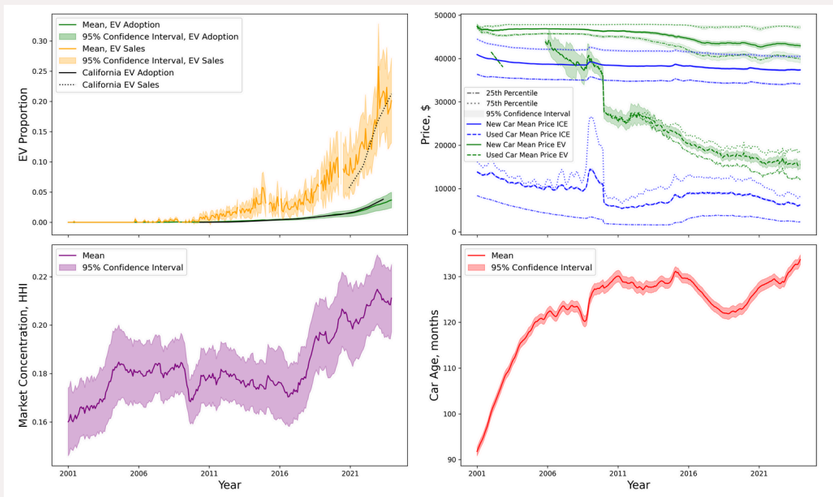
- What policy combinations that meet adoption targets can best balance the trade-offs between costs, consumer utility, and emissions?

MODEL STRUCTURE



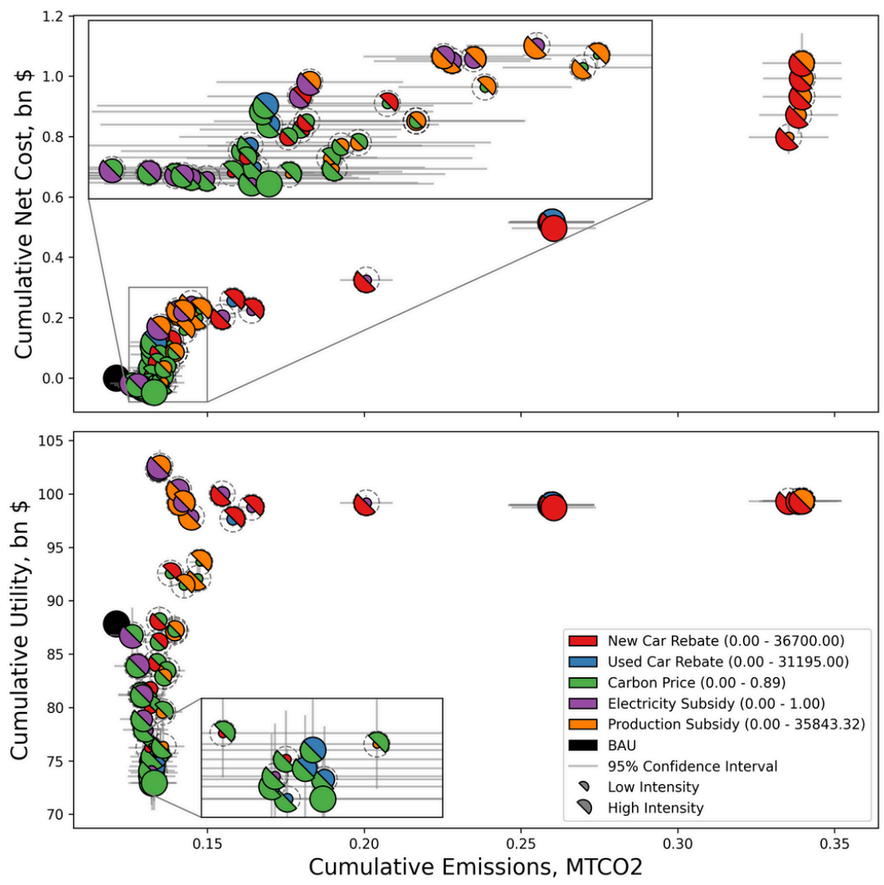
MODEL

- Consumers choose between keeping, buying new, or buying used cars.
- Firms innovate by exploring a multi-dimensional NK landscape and price-based profit maximization.
- Social imitation determines willingness to consider EVs as an alternative to ICEVs.
- Five policies are tested until 2035: Carbon price, Subsidy on EV charging electricity, Rebate for new EVs, Rebate for used EVs, and Production subsidy.
- Policies aim to achieve a 95% EV car fleet improving on the business-as-usual (BAU) scenario of 40% uptake.

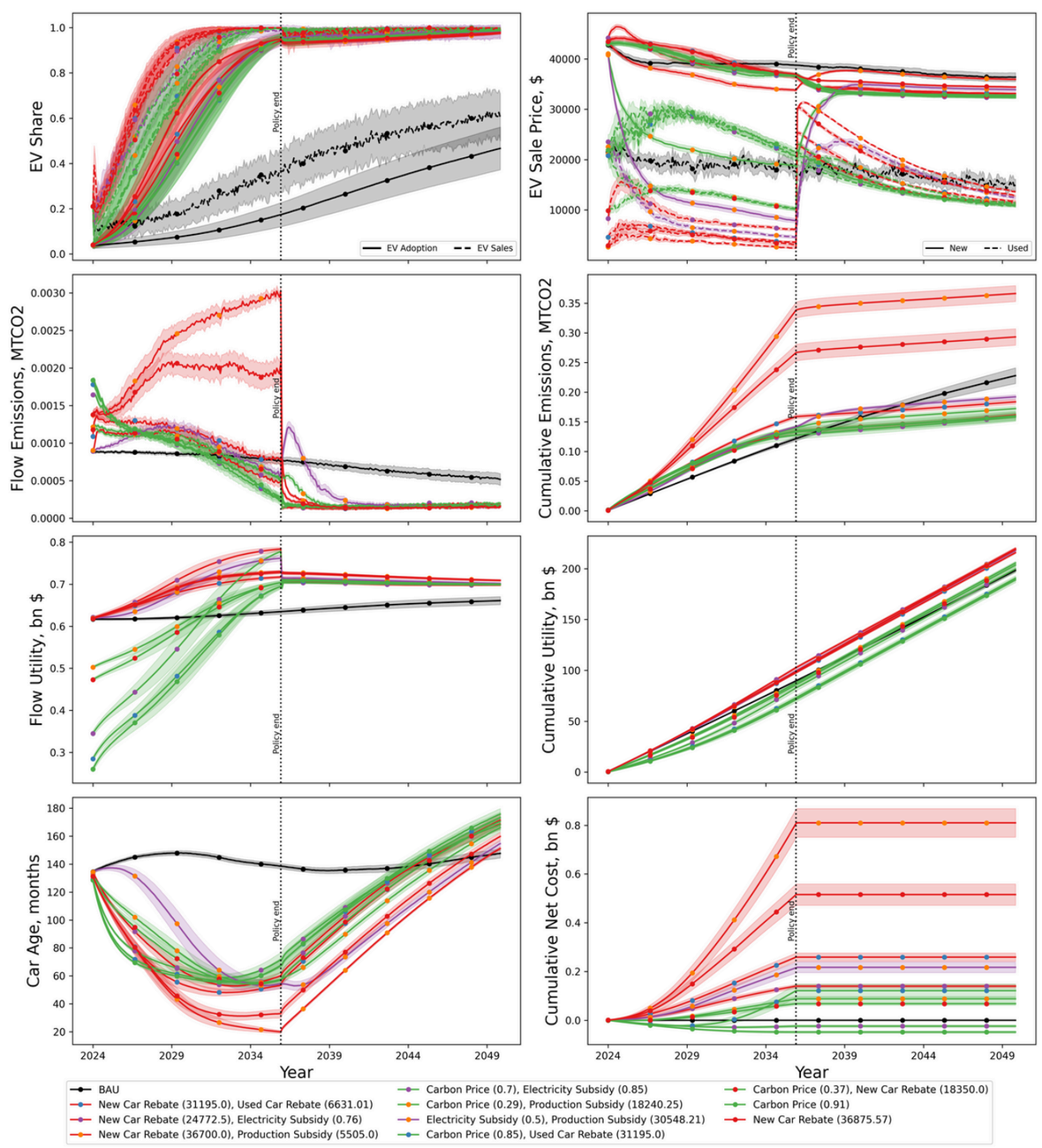


Model Parameters calibrated to match California EV data from 2010-23 using time series data and policies from 2001-23

RESULTS



Policies pairs outperform single policies (carbon price and new car rebate) over multiple criteria, but still have tradeoffs relative to BAU



Test low-intensity policy pairs, remove the policy instruments in 2035 and run the projection until 2050 to test policy stability.

CONCLUSIONS

- Single policies such as carbon pricing and New car rebates can achieve rapid transition but with either larger sacrifices in utility or high emissions rebound from manufacturing.
- Combinations of moderate policies are more feasible and effective, alleviating some of the worst trade-offs, such as subsidies to EV electricity use increasing utility.
- After policies end, high EV adoption is stable and beneficial, but there is a need to tackle car dependence overall, not just vehicle technology.

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